

Blind Hiring – Exploratory / Robustness Results

0.1 Alternative A.1 (HR only): Candidate-Level Selection Probability Model with `sc_advance`

The per- k loop in §4.2–§4.6 suffers from extreme discrete-sampling variance at the boundaries (at $k=1$ a single tie-break injects pure noise into the dependent variable; at $k=18$ the pools collapse to identical sets). For HR, we have a natural binary — `sc_advance`, the actual decision to advance a candidate to the engineering stage — which lets us re-estimate the same selection question on the full candidate-evaluator panel without any k threshold:

$$\underbrace{\text{sc_advance}_{ij}}_{\text{HR's advance decision}} \sim \text{TrueCoding}_i \times \mathbb{1}[\text{Blind}]_j \times \text{Moderator}_j \mid \text{Respondent FE, cluster on Respondent}.$$

Why HR only here: `sc_advance` is HR's natural single-decision move forward'' binary; engineers do not produce a comparable single-shot binary on the firm side --- they instead deliver an `\emph{ordering}` over the 18 resumes, from which any binary is necessarily a top- k thresholding (which is exactly the §4.2 design we are trying to escape). For the engineer side we therefore push the analogous does blinding strengthen the link between true ability and selection'' question into a k -loop variant in §4.8 below.

With respondent FE, all between-respondent terms (`treat::blind`, the moderator main effect, and their two-way interaction) are absorbed; identification of the within-respondent slopes and their cross-products comes from variation in true coding ability across the 18 resumes a respondent rates. The substantive coefficients of interest are:

- **True Coding × Blind** — the central efficiency claim. > 0 means blinding tightens the link between objective ability and being advanced.
- **True Coding × Blind × M** — whether blinding's tightening of that link itself depends on the moderator (DEI orientation, firm DEI policy, or inter-departmental DEI misalignment).

0.2 Alternative A.2 (k -loop): Selection Probability for Both Roles, Looping over k

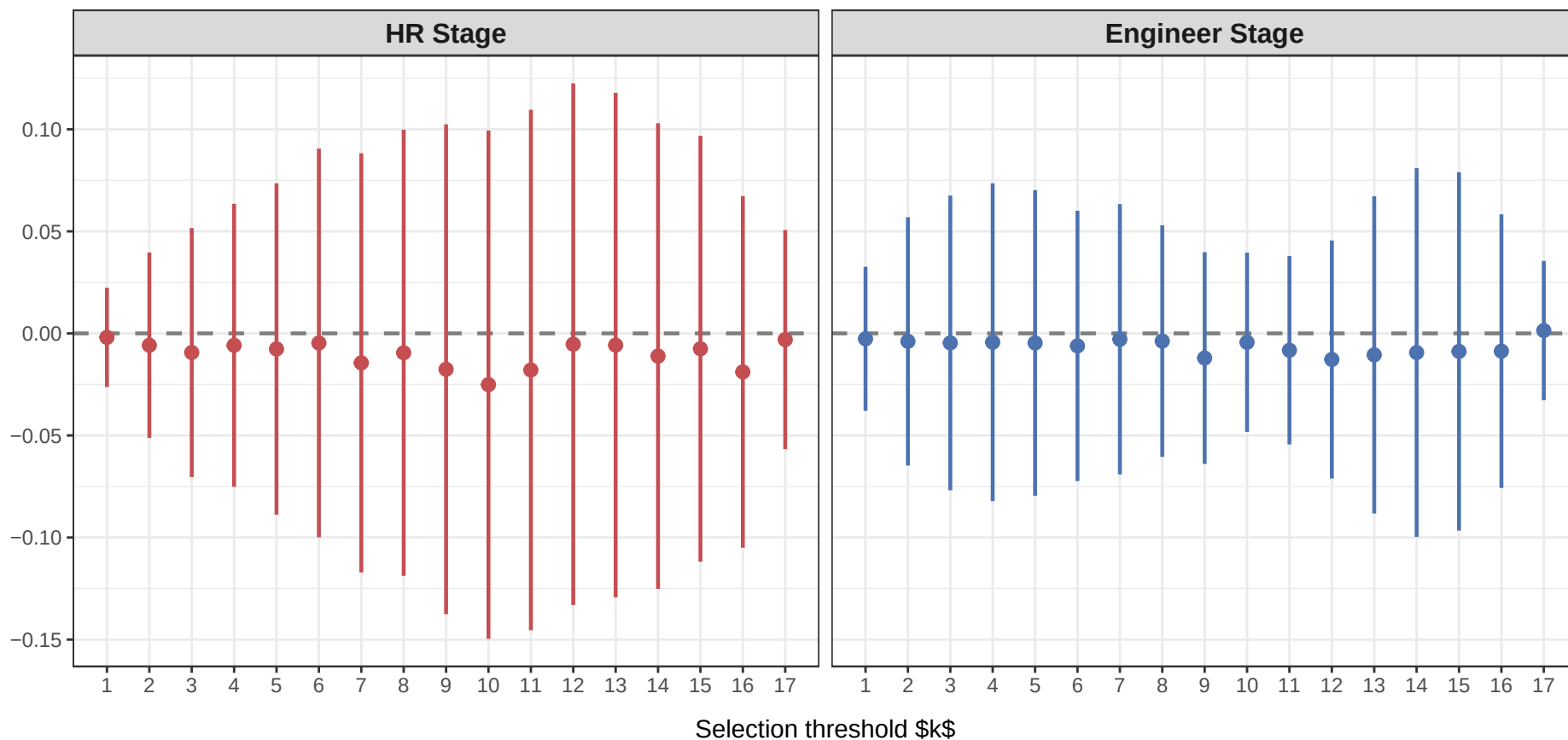
Relationship to §4.2–§4.6: Because engineers do not give a single advance/no-advance decision, the analog of §4.7 for engineers requires picking a binary — and the only natural ones are the k -thresholded indicators `hr_passk` / `eng_topk`. Doing so produces what is essentially the LHS↔RHS *dual* of §4.2: where §4.2 conditions on the selected pool and asks *\emph{Is the average true coding score of the selected k higher under blinding?}*, ' §4.8 instead uses the full candidate-evaluator panel and asks *Is true coding ability a stronger predictor of being one of the selected k under blinding?*' Mathematically the two are not identical (different estimands and different units), but they are tightly linked: if blinding raises the average true score of the chosen k , it must be (approximately) because blinding has steepened the within-respondent gradient of selection probability on true ability. The advantage of the dual specification is that the panel sample size at every k is the full $N \approx 5,238$ candidate-evaluator observations, which compresses the standard errors that explode in §4.2 at low k .

$$\underbrace{\text{Selected}_{ij}^{(k)}}_{\substack{\text{HR: hr_pass}_k \\ \text{Eng: eng_top}_k}} \sim \text{TrueCoding}_i \times \mathbb{1}[\text{Blind}]_j \mid \text{Respondent FE, cluster on Respondent}, \quad k = 1, \dots, 18.$$

The plotted quantity is the per- k `true_coding_z × treat::blind` coefficient: the change in the within-respondent linear-probability slope of selection on true coding ability when going from non-blind to blind review. Positive means blinding tightens the ability-selection link. Bands are 95% CIs from clustered SEs on respondent.

Slope shift in P(selected at top-k) per 1 SD true coding ability (blind – nonblind)

Per-k change in the within-respondent LPM slope of selection on true coding ability under blinding (treat::blind x true_coding_z coefficient)



LPM with respondent FE + display-order (r_do) FE; two-way clustered SE on (respondent, resume). Resume-content FE intentionally not absorbed (would kill the true_coding_z slope). Po:

0.3 Alternative B: Evaluator-Level Rank Correlation (“Accuracy Score”)

Both prior alternatives still operate on the resume-evaluator long panel. Alternative B collapses that panel to one number per evaluator and asks a different but complementary question: across the 18 resumes a single evaluator rated, *how well does their overall ranking of those 18 resumes line up with the objective ranking by lab coding score?* For each respondent r we compute

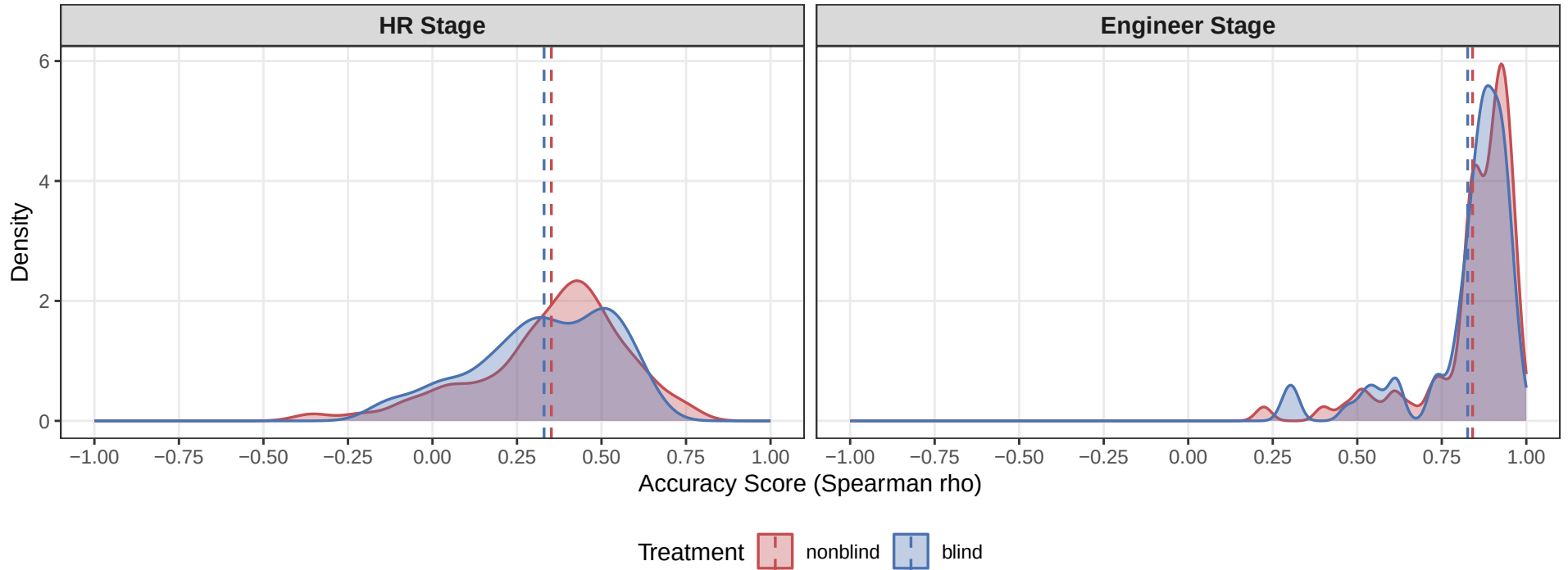
$$\rho_r = \text{cor}\left(\text{rank}(\widehat{\text{sc_overall}}_{r,\cdot}), \text{rank}(\text{overall_score}_{\cdot})\right) \in [-1, +1],$$

the Spearman rank correlation between the evaluator’s overall ratings (`sc_overall_z`) and the resumes’ objective lab coding scores (`overall_score`). $\rho_r=+1$ means the evaluator’s ranking is identical to the objective ranking; $\rho_r=0$ means pure noise; $\rho_r=-1$ means perfectly inverted. Spearman is rank-based, so the choice between raw and standardized versions of either input is irrelevant. We then regress the Fisher-z transform $\tilde{\rho}_r = \tanh^{-1}(\rho_r)$ (variance-stabilized, unbounded) on treatment and a moderator:

$$\tilde{\rho}_r \sim \mathbb{1}[\text{Blind}]_r \times \text{Moderator}_r + \text{controls}, \quad \text{HC1 robust SEs.}$$

Each observation is one evaluator (one row), so no clustering is needed — independence is by design. **Blind** positive $\Rightarrow$ blinding raises rank-alignment with truth on average; **Blind** $\times$ M nonzero $\Rightarrow$ that improvement varies with the moderator.

Distribution of evaluator-level Accuracy Score (Spearman rho between `sc_overall_z` and `overall_score` across 18 resumes)



Dashed lines: arm-specific means. $\rho > 0$ = evaluator’s overall ranking aligns with objective coding ability; $\rho < 0$ = anti-aligned.

Table 1: Alternative B — Evaluator-Level Accuracy Score: Fisher-z transformed Spearman rank correlation between the evaluator’s overall ratings and resumes’ objective lab coding scores across the 18 resumes they rated. OLS with HC1 robust SEs in parentheses; one observation per evaluator (no clustering). Resume-content / Display Position / Respondent FEs not applicable (respondent-level regression). Standard demographic Respondent Characteristics (resp_gender, resp_race, s2_educ, s2_income) absorbed in both HR and Eng specs. Eng additionally absorbs coding-language FEs (resp_python, resp_java), which are Eng-only survey items.

	HR				Engineer			
	(1) Pool	(2) DEI	(3) FirmTgt	(4) Misalign	(5) Pool	(6) DEI	(7) FirmTgt	(8) Misalign
Blind	0.002 (0.025)	−0.001 (0.025)	−0.002 (0.027)	−0.011 (0.029)	−0.065 (0.041)	−0.067* (0.041)	−0.046 (0.042)	−0.068 (0.047)
M (moderator)		0.027 (0.018)	−0.034 (0.046)	0.042** (0.019)		−0.095*** (0.034)	−0.138 (0.127)	−0.053 (0.034)
Blind × M		−0.053** (0.023)	0.008 (0.067)	−0.016 (0.025)		0.052 (0.039)	−0.129 (0.161)	−0.016 (0.054)
Resp. age	0.003** (0.001)	0.003** (0.001)	0.003** (0.001)	0.001 (0.001)	−0.004 (0.004)	−0.006 (0.004)	−0.003 (0.004)	−0.004 (0.004)
Display Position FE	×	×	×	×	×	×	×	×
Resume ID FE	×	×	×	×	×	×	×	×
Resume - Coding Test Version FE	✓	✓	✓	✓	✓	✓	✓	✓
Respondent ID FE	×	×	×	×	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓	✓	✓	✓	✓
Obs.	411	411	411	339	462	462	462	354
R^2	0.139	0.148	0.141	0.182	0.061	0.085	0.080	0.116
Adj. R^2	0.093	0.098	0.090	0.123	0.014	0.035	0.029	0.052

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

1 Within-Session Demographic Calibration (Robustness)

1.1 Spec 1: Does Demographic-Conditional Anchoring Exist?

Question: Does the running mean of objective coding ability among *same-demographic* candidates seen earlier in the session predict the current candidate’s evaluation score? ‘lag_diff_z’ (other-demographic running mean) is the placebo.

	HR (nonblind)	Eng (nonblind)	HR (blind, placebo)	Eng (blind, placebo)
Same-Demo Prior Mean (Z)	0.059 (0.050)	0.080*** (0.025)	−0.016 (0.040)	0.028 (0.030)
Other-Demo Prior Mean (Z, placebo)	0.002 (0.042)	0.033 (0.023)	0.041 (0.038)	−0.027 (0.025)
Observations	785	878	835	950
R ²	0.444	0.792	0.478	0.735
Adj. R ²	0.407	0.779	0.445	0.721

* p < 0.1, ** p < 0.05, *** p < 0.01

Standard errors clustered by respondent. Sample: rows with *r_dogeq2* and at least one prior of the indicated demographic class.

1.2 Spec 3: Are the §1.1 ATEs Robust to Lag Controls?

Question: If we add the Spec-1 lag controls to the §1.1 race-gender regression (full sample), do the demographic ATEs shrink? Both columns per role run on the same restricted sample (rows with both lags non-NA) so the comparison is purely about specification.

	Resume-Screening Phase		Technical-Evaluation Phase	
	HR (base)	HR (with lag)	Eng (base)	Eng (with lag)
White Male	0.043 (0.073)	0.045 (0.073)	0.045 (0.039)	0.037 (0.039)
White Female	0.019 (0.060)	0.016 (0.061)	0.019 (0.042)	0.011 (0.041)
Asian Male	0.037 (0.082)	0.037 (0.083)	-0.001 (0.033)	-0.002 (0.033)
Asian Female	-0.081 (0.063)	-0.082 (0.063)	-0.021 (0.035)	-0.024 (0.035)
Hispanic Male	-0.113 (0.069)	-0.113 (0.069)	-0.010 (0.037)	-0.011 (0.038)
Hispanic Female	-0.021 (0.075)	-0.021 (0.075)	0.010 (0.046)	0.003 (0.046)
Same-Demo Prior Mean (Z)		-0.025 (0.032)		-0.051** (0.020)
Other-Demo Prior Mean (Z)		-0.018 (0.028)		-0.003 (0.017)
Observations	1620	1620	1828	1828
R ²	0.447	0.448	0.756	0.757
Adj. R ²	0.428	0.428	0.748	0.749

* p < 0.1, ** p < 0.05, *** p < 0.01

Coefficients sign-flipped post-fit so positive = effect of blinding raises the score for that demographic. Standard errors clustered by respondent. Same restricted sample for base and lag-controlled columns within each role.